

Order Flow Imbalance Features: Conceptual Questions

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1. Motivation for Measuring OFI at Multiple Depth Levels

Order Flow Imbalance (OFI) quantifies the net pressure from limit order book (LOB) events—additions, deletions, and executions—on the price. Measuring OFI only at the best bid and ask prices (best-level OFI) overlooks the latent liquidity and trading intentions in deeper levels of the book. Research by Xu et al. (2018) and Kolm et al. (2023) highlights that the depth at levels 2 to 10 can exceed that of level 1, suggesting significant market participant activity and potential price-relevant information is embedded deeper in the book.

Hence, multi-level OFI (as defined by Eq. (2) in the paper by Zhang et al.) captures a richer representation of market dynamics. This approach acknowledges that informed traders may strategically place or cancel orders away from the best level to minimize market impact. Incorporating deeper levels improves the robustness and stability of predictive models, especially when combined into an *Integrated OFI* through PCA (Principal Component Analysis), which reduces noise and redundancy from high-dimensional LOB data [1].

2. Why Lasso Regression is Preferred Over OLS for Estimating Cross-Impact

Estimating cross-asset impact involves regressing the return of one asset on the OFI of many others. In this setting, using Ordinary Least Squares (OLS) becomes problematic due to high dimensionality ($N \approx 100$) and multicollinearity among predictors. The condition number of the OFI matrix often violates the assumptions necessary for a stable OLS solution.

Lasso regression mitigates these issues by performing both coefficient shrinkage and variable selection. It imposes an l_1 -norm penalty that drives many coefficients to zero, yielding sparse, interpretable models. This is crucial when only a few assets have meaningful cross-impacts on any given stock. According to Zhang et al., Lasso not only enhances statistical significance but also aligns with the sparsity assumption empirically observed in financial networks [1].

3. Why OFI is a Better Predictor of Short-Term Returns Than Trade Volume

While trade volume aggregates executed orders, OFI provides a directional measure of market pressure across all order types: limit orders, market orders, and cancellations. Volume lacks context—it does not differentiate between buying and selling pressure, nor does it reflect liquidity dynamics in the LOB.

OFI, in contrast, encapsulates *intentional* order flow imbalances that directly affect short-term price formation. Empirical evidence from Cont et al. (2014) and Zhang et al. (2024) shows that OFI explains a significantly higher proportion of the variance in short-term returns (adjusted R^2 exceeding 87% with integrated OFI) than trade volume, due to its sensitivity to transient order book dynamics [1, 2].

References

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